

Data Warehousing and FTP Serving

In this article, we will set up a FTP server, and then discuss FreeNAS, an operating system based on FreeBSD, which helps set up a data storage server.

More often than not, data comes as files. Efficiently storing these files is a major headache, and then making those files available to the general public or to specific authenticated users is even more so. In this article, we will set up a FTP server, and then discuss FreeNAS, an operating system based on FreeBSD, which helps set up a data storage server.

Section A: The FTP server

Setting up the FTP server requires some out-of-the ordinary steps if you don't have RPMForge. Those steps are—installing RPMForge! After you are done, open up a terminal prompt and type in the following:

```
yum install proftpd-mysql
```

That's all for the installation part. Before we move on to configuration, we need to set up ProFTPD to run automatically at system startup. To do that, execute the following:

```
chkconfig --level 345 proftpd on
```

And for the configuration, open up the file `/etc/proftpd.conf.in` in a text editor, and read on.

The first task is enabling Anonymous FTP. If you want to set up a FTP server to serve downloads to the general public (like ftp.gnu.org), you need this part. The 'Anonymous' section is in the config file at the very end, but it's commented out. Uncomment the whole section. There's another small task; the config file has an invalid directive called 'DisplayFirstChdir'. Run the following Perl command to correct it:

```
perl -pi -e 's/DisplayFirstChdir/DisplayChdir/' /etc/proftpd.conf
```

Another bit of work involves adding a directive. The directive will enable FTP access for accounts that do not have



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a shell. In the 'Anonymous' section, add the following line:

```
RequireValidShell    Off
```

You're done! Just type in...

```
service proftpd start
```

...at a terminal and you're good to go!

Section B: Data warehousing

To run your own download server and host a big repository of source code or even company documents that would take up terabytes of storage space, repeatedly adding more hard disks to a single server would be impractical. In fact, it would be suicidal. So many pieces of hardware inside a single computer put a lot of load on the MoBo and the SMPS. Moreover, how many hard drives can you possibly attach to a single MoBo?

There is a solution, and it's called Network Attached Storage (NAS). A grid of computers interconnected with the Ethernet and running NAS servers is called a SAN, or Storage Area Network. But here's the good news: NAS comes cheap.

To have a NAS machine, you don't need very good hardware. A 600 MHz Transmeta Crusoe or Efficeon will suffice. Hey, you can even have ARM boxes. And in case you think you need something more, out here in Kolkata, eSys sells a Mini-ITX 'System In A Box', consisting of a 1.6 GHz Intel Atom processor mounted on an Intel 82945G Express Chipset-based motherboard, for a little over Rs 2,000. That's a good deal. As for RAM, you need about 512 MB for a decent performance.